DATA SCIENCE PROJECT · CSE

Airport Monitoring System

A data-driven approach to air traffic flow, delay prediction, runway safety and passenger-flow monitoring

Shivam Kaul

Enrollment No: S25CSEU0918

B.Tech · Computer Science Engineering

Agenda

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Introduction

An Airport Monitoring System uses live operational data — flight schedules, radar feeds, weather, and ground-sensor streams — to track, analyse and predict everything happening across an airport in real time.

Applying data science turns this raw operational data into early warnings, accurate delay forecasts and safer runway operations — instead of reactive, manual monitoring.

37M+

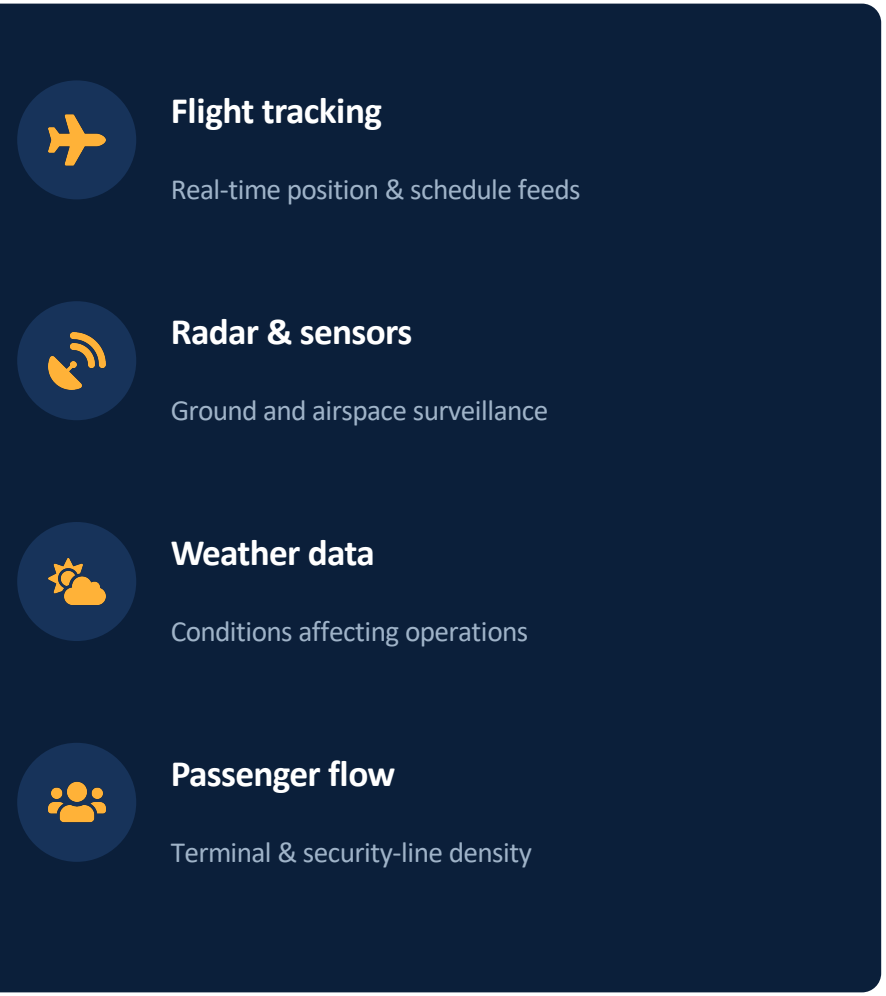
Flights tracked globally / year

~20%

Flights facing avoidable delays

24/7

Continuous monitoring need



Problem Statement

Airports run on tightly coupled systems — a small disruption in one area cascades into delays, congestion and safety risk across the whole network.

Air Traffic Congestion

Growing flight volumes strain runway and airspace capacity, causing bottlenecks during peak hours.

Flight Delays & Cascading Effects

A single delayed aircraft disrupts crew schedules, gate slots and connecting flights downstream.

Runway & Ground Safety

Manual monitoring struggles to catch incursions, near-misses or equipment faults in real time.

Weather-Related Disruptions

Sudden weather changes are hard to factor into scheduling and rerouting decisions quickly.

Passenger Flow & Congestion

Security lines, boarding gates and terminals get overcrowded without predictive crowd data.

Fragmented, Reactive Systems

Data lives in silos (ATC, ground ops, weather); problems are spotted after they occur, not before.

Objectives



Predict flight delays

Build models that forecast delay likelihood and duration using historical + live flight data.



Monitor air traffic flow

Analyse airspace and runway occupancy to flag congestion before it builds up.



Detect anomalies early

Flag unusual patterns in ground movement, weather or scheduling that signal risk.



Model passenger flow

Estimate terminal and security-line density to support better crowd management.



Support runway safety

Surface real-time alerts for potential incursions or capacity violations.



Unify fragmented data

Bring ATC, weather, ground-ops and passenger data into one monitoring pipeline.

Data Sources & Methodology

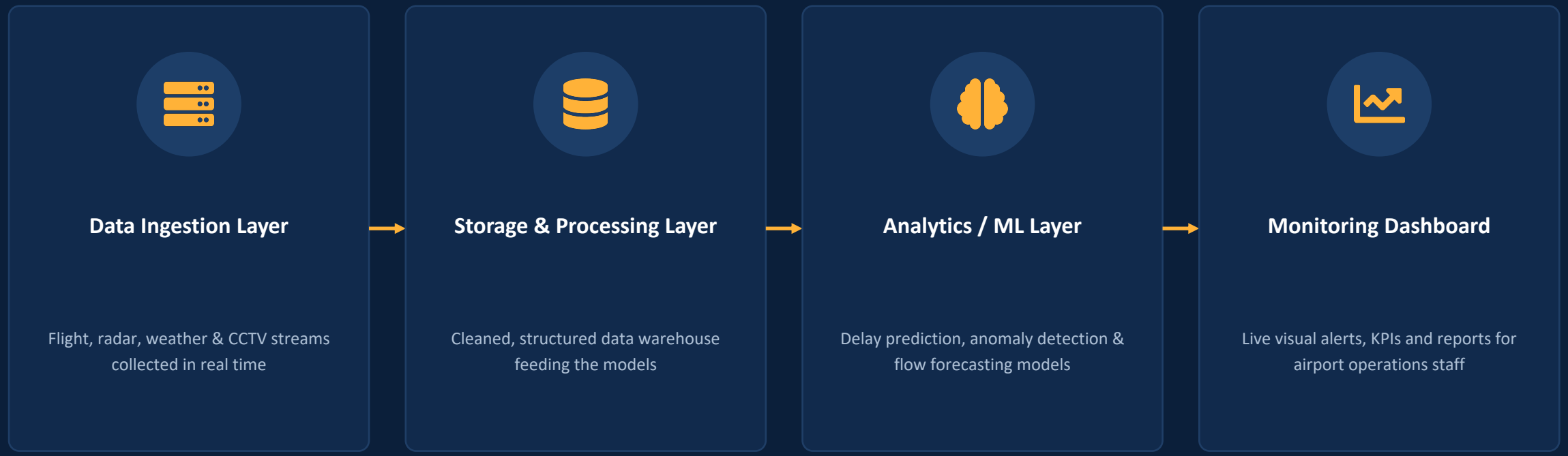
Data Sources

- Flight schedule & OTP records (departure/arrival, aircraft type, airline)
- Weather feeds (visibility, wind speed, precipitation, temperature)
- ATC / radar & ADS-B tracking data (position, altitude, speed)
- CCTV / ground-sensor feeds for runway & terminal monitoring

Methodology Pipeline

1. **Data Collection**
2. **Cleaning & Preprocessing**
3. **Feature Engineering**
4. **Model Training**
5. **Evaluation**
6. **Real-time Deployment**

System Architecture



Modelling & Techniques



Regression Models

Predict delay duration (Linear/XGBoost Regression) from schedule + weather features.



Classification Models

Random Forest / Gradient Boosting to classify flights as on-time, delayed or cancelled.



Anomaly Detection

Isolation Forest / clustering to flag unusual ground-traffic or sensor patterns.



Time-Series Forecasting

LSTM / ARIMA models to forecast runway and airspace congestion by hour.



Computer Vision

CNN-based detection on CCTV feeds for runway incursions & obstacle detection.

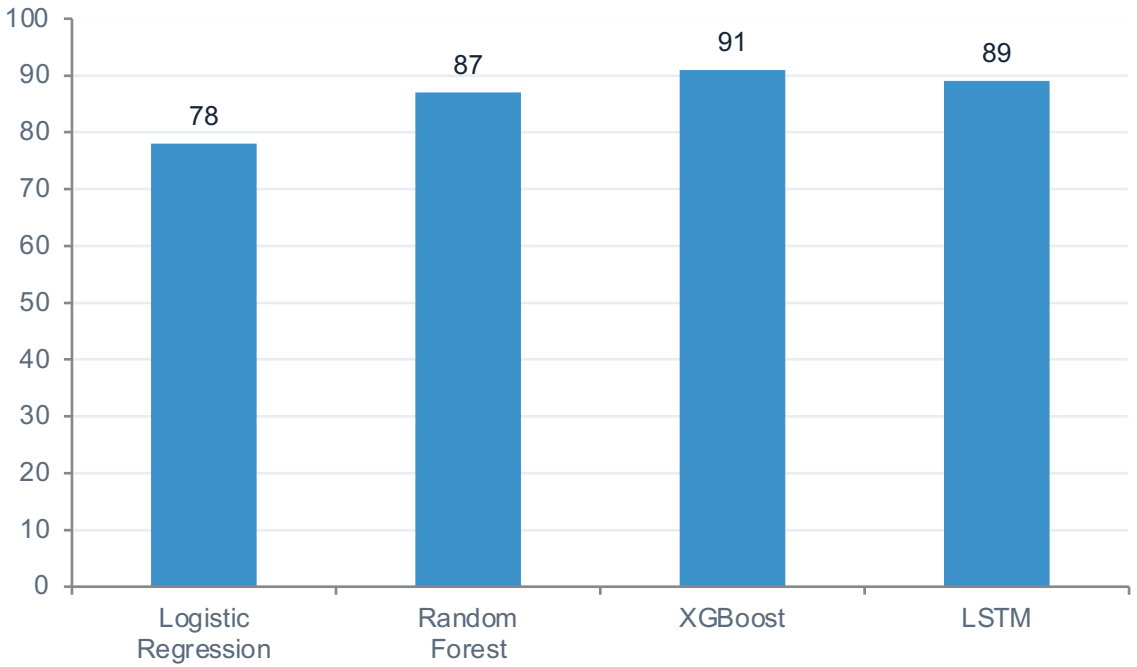


Clustering & Flow Analysis

K-Means to group traffic/passenger patterns and identify peak congestion zones.

Expected Results & Evaluation

Model Performance Comparison (Accuracy %)



Key Evaluation Metrics

Accuracy / F1-score

for delay classification models

RMSE / MAE

for delay-duration regression

Precision & recall

for anomaly / incursion alerts

Latency

of real-time dashboard updates

Expected outcome: fewer cascading delays, earlier safety alerts, and a single live view of airport operations.

Challenges & Future Scope

Challenges

- Access to real-time, high-quality airport data for training
- Integrating multiple heterogeneous data sources (ATC, weather, CCTV)
- Ensuring low-latency predictions for real-time alerts
- Handling rare-event data for accident/incursion detection
- Data privacy & security for surveillance feeds

Future Scope

- Integrate with live ADS-B / ATC feeds for real deployment
- Add computer-vision runway-incursion detection via live CCTV
- Extend to multi-airport, network-wide congestion prediction
- Deploy as a real-time dashboard with automated alerts
- Use reinforcement learning for dynamic gate/slot allocation

Conclusion

A data-science-driven Airport Monitoring System can turn scattered flight, weather, radar and CCTV data into early, actionable insight — reducing delays, easing congestion and strengthening runway safety through continuous, predictive monitoring rather than reactive response.



Thank You

Questions & Discussion

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